A NON-EFFECTIVE FINITE EQUIVALENCE RELATION ON A FINITE SCHEME OVER A FIELD, AFTER A QUESTION OF GROTHENDIECK

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Abstract. In 1961 Grothendieck asked whether a closed equivalence relation on a finite scheme over a field, with finite but not necessarily flat projections, is always effective, and in a lecture at Buffalo in 1973 he still held the question open. We show that the answer is no over every field. Restricting Raicu's non-effective relation on the affine plane to a finite subscheme of length 27 cut out by powers of invariant functions gives a closed equivalence relation on the spectrum of a local Artin algebra with residue field k which is not effective: its ideal is strictly larger, in degrees 5 and 6, than the ideal of $X \times_Y X$. The comparison is a finite computation in linear algebra; we carry it out over $\mathbb{Q}$, record the primes that divide a pivot, and check those characteristics separately, so that it holds over every field.

1. The question

Let k be a field, $X = \operatorname{Spec} A$ a finite k-scheme and $R \subset X \times_k X$ a closed subscheme whose points with values in every k-algebra form an equivalence relation. Its projections $R \rightrightarrows X$ are finite, but they need not be flat. The ring of invariants

$$B = A^R = \{\, a \in A : a \otimes 1 - 1 \otimes a \in I_R \,\},$$

with $I_R \subset A \otimes_k A$ the ideal of R, is a finite k-algebra, and $Y = \operatorname{Spec} B$ is the quotient of X by R in the category of schemes. The relation is *effective* when the morphism $R \to X \times_Y X$ is an isomorphism.

In the third of his talks on the construction of schemes, after the flat case, Grothendieck wrote that without flatness he did not know whether a finite equivalence relation is effective. He reduced the question to the spectrum of a local Artin ring with algebraically closed residue field, and said that it was open even over a field [2, §2, p. 212-05]. Twelve years later, at the end of a lecture of his algebraic geometry course at the State University of New York at Buffalo, on 11 July 1973, a listener asked for open problems and he gave this one: he said it was not known, that he expected the answer to be negative, and that it was very elementary [1]. The flat case is SGA 3, exp. V [7].

Without flatness, two kinds of negative results are known. Venken showed that descent of flat modules along a finite morphism of local Artin rings need not be effective [8], and Philippe studied descent along unramified morphisms [4]; these concern modules, not the relation itself. In the appendix to Kollár's paper on quotients by finite equivalence relations, Raicu gave finite equivalence relations on $\mathbb{A}^2$ which are not effective [3, §50], while affine toric equivalence relations are effective [5]. We have not found the case of a finite scheme over a field treated in print, in either direction.

Theorem 1. *Let k be any field. There is a local Artin k-algebra A of dimension* 27*, with residue field k, and a closed equivalence relation R on* $\operatorname{Spec} A$*, with finite projections, which is not effective.*

Over a field the answer is therefore negative, as Grothendieck expected, and the more general question, for X finite over a noetherian base, is answered by the same example.

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2. Effectivity as linear algebra

With the notation of the question, $X \times_Y X = \operatorname{Spec}(A \otimes_B A)$ and $A \otimes_B A = (A \otimes_k A)/J_B$, where J_B is the ideal generated by the $b \otimes 1 - 1 \otimes b$, $b \in B$. By definition of B, $J_B \subseteq I_R$, and $R \to X \times_Y X$ is the closed immersion defined by I_R/J_B. So R is effective if and only if $I_R = J_B$, a comparison of two subspaces of the finite-dimensional space $A \otimes_k A$.

When A and I_R are graded, B is a graded subalgebra and both ideals are graded, so the comparison can be made degree by degree. A homogeneous invariant of degree e contributes to J_B only in degrees $\geq e$; to compare the two ideals in degree d it suffices to know the invariants of degree at most d.

3. Raicu's relation and its restriction

Write $x = (x_1, x_2)$ and $y = (y_1, y_2)$ for the coordinates of $\mathbb{A}^2 \times \mathbb{A}^2$. Raicu's relation [3, §50] is the closed subscheme $R \subset \mathbb{A}^2 \times \mathbb{A}^2$ with ideal

$$I = (x_1^2 - y_1^2,\ x_1x_2 - x_2^2 - y_1y_2 + y_2^2,\ x_2^3 - y_2^3,\ (x_1y_2 - x_2y_1)\, y_2^3) \subset k[x_1, x_2, y_1, y_2].$$

It is a finite equivalence relation on $\mathbb{A}^2$, defined over $\mathbb{Z}$, and it is not effective after base change to any field; the class of $(x_1y_2 - x_2y_1)\, y_2^3$ is the obstruction. The functions

$$f_1 = x_1^2, \qquad f_2 = x_1x_2 - x_2^2, \qquad f_3 = x_2^3$$

are invariant, since $f_i(x) - f_i(y)$ is one of the first three generators of I. Let

$$Z = V(f_1^3, f_2^3, f_3^2) \subset \mathbb{A}^2, \qquad A = k[x_1, x_2]/(f_1^3, f_2^3, f_3^2).$$

Since $f_1^3 = x_1^6$ and $f_3^2 = x_2^6$, Z is supported at the origin, and A is a local Artin algebra with residue field k; a count of monomials shows that $\dim_k A = 27$. Let $R_Z = R \cap (Z \times Z)$, the closed subscheme of $Z \times Z$ with ideal I_{R_Z}, the image of I in $A \otimes_k A$.

Lemma 2. *R_Z is a closed equivalence relation on Z with finite projections.*

Proof. For every k-algebra T, $R_Z(T) = R(T) \cap (Z(T) \times Z(T))$ is the restriction of an equivalence relation on $\mathbb{A}^2(T)$ to the subset $Z(T)$, hence an equivalence relation. R_Z is closed in $Z \times Z$, which is finite over Z by either projection. □

The restriction to Z is what lets a question about $\mathbb{A}^2$ be asked about a finite scheme. It is not formal that non-effectivity survives it: the ring of invariants of R_Z is computed on A, not on $k[x_1, x_2]$, and it can be larger than the image of the invariants on $\mathbb{A}^2$, so that J_B can grow. The next section shows that it does not grow enough.

4. The computation

All the ideals above are homogeneous for the standard grading. In $k[x, y] = k[x_1, x_2, y_1, y_2]$ put $K = (f_1^3, f_2^3, f_3^2)(x) + (f_1^3, f_2^3, f_3^2)(y)$, so that $A \otimes_k A = k[x, y]/K$. The ideal of R_Z lifts to $I + K$, and J_B lifts to $J + K$, where J is generated by the $b(x) - b(y)$ for b running over homogeneous lifts of a basis of B.

Proposition 3. *Over $\mathbb{Q}$ and over every finite prime field, the invariants of R_Z of degree at most 7 give 22 independent differences $b(x) - b(y)$ modulo $I + K$, and in the degrees $d \leq 14$ the two ideals differ exactly in*

$$\dim_k(I + K)_5 = 44 > 43 = \dim_k(J + K)_5, \qquad \dim_k(I + K)_6 = 77 > 75 = \dim_k(J + K)_6.$$

By the remark at the end of Section 2, invariants of degree at least 8 do not change $J + K$ in degrees 5 and 6, so the strict inequality in degree 5 proves that R_Z is not effective. A witness is the generator $(x_1y_2 - x_2y_1)\, y_2^3$ of I, Raicu's obstruction: it lies in $(I + K)_5$ and not in $(J + K)_5$, as we checked over $\mathbb{Q}$, $\mathbb{F}_2$, $\mathbb{F}_3$ and $\mathbb{F}_5$.

The proposition is a finite computation: the degree-d parts are spanned by monomial multiples of the generators, the invariants of degree d are the kernel of a linear map on the forms of degree d in x, and the dimensions are ranks. The script `full.py` of the directory

`computations/effective/raicu` of the repository [6] performs it by Gaussian elimination over $\mathbb{Q}$ or over $\mathbb{F}_p$.

Proof that it holds over every field. A field k contains $\mathbb{Q}$ or some $\mathbb{F}_p$, and all the data are defined over the prime field: ranks and kernels of matrices with entries in the prime field do not change under extension of scalars. It remains to cover every prime field.

Over $\mathbb{Q}$ the elimination divides only by pivots. Each coefficient it produces is an integer combination of the input coefficients, all integers, divided by a product of pivots. If a prime p divides neither the numerator nor the denominator of any pivot, the same sequence of operations is defined over $\mathbb{F}_p$, with the same pivot positions, and gives the same ranks and the same kernels. The script `certify.py` reruns the computation over $\mathbb{Q}$ and records the prime factors of all pivots; they are $2, 3, 5, 7, 11, 13, 41, 61, 67$ and 661. For every other prime the proposition over $\mathbb{F}_p$ follows from the one over $\mathbb{Q}$. For these ten primes, and for all primes up to 101 and for 1009 and 10007 as a cross-check, `full.py` was run directly over $\mathbb{F}_p$, with the same result. □

Remark 4. Why Z and not something smaller. Before this example, a random search over 48 small monomial algebras over $\mathbb{F}_2$ and $\mathbb{F}_3$ found no non-effective relation. Raicu's theorem that affine toric equivalence relations are effective [5] suggests why: monomial relations are close to the toric case. We did not look for the smallest finite subscheme on which the obstruction survives.

Remark 5. Grothendieck reduced the question to an algebraically closed residue field; the example has residue field k, which may be chosen algebraically closed. It is also graded, which the question does not require.

What is not done. The computation is not formalised: a Lean proof would need either a certified Gaussian elimination over $\mathbb{F}_3$ and $\mathbb{Q}$ or a Gröbner-basis certificate, and the pivot argument above. The example has not been checked by a specialist of descent.

Use of language models. The question was located in the recording of the Buffalo lecture and in [2] with language models (Claude Opus 5.5, Anthropic), which also found Raicu's example, proposed the restriction to Z, wrote the scripts and a first draft of this text, under the author's direction. The proposition rests on the scripts, which anyone can rerun.

References

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